

Interactive Physics Simulations

Interactive Learning Platform

Quantum Mechanics Simulations & Educational Tools

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1. Project Overview

This project is an interactive web-based educational platform designed to help students and educators explore fundamental concepts in quantum mechanics and chemistry. The platform provides a collection of physics simulations and visualizations that bring theoretical concepts to life through interactive experimentation.

Project Goals:

- Provide interactive, web-based tools for learning quantum mechanics
- Visualize complex physical phenomena in an accessible way
- Integrate lecture materials with hands-on simulation experiences
- Enable students to experiment with parameters and observe real-time effects

Target Audience:

Physics and chemistry students at all levels, as well as educators and learners seeking interactive teaching and learning materials for quantum mechanics and chemistry courses.

2. Technologies Used

Technology	Version	Purpose
TypeScript	5.9.3	Primary programming language for type-safe development
Vite	7.2.4	Fast build tool and development server
Three.js	0.182.0	3D visualization library for atomic orbitals and molecular graphics
Chart.js	4.5.1	2D plotting and graphing for energy diagrams and data visualization
KaTeX	0.16.28	Mathematical formula rendering and typesetting
Marked	17.0.2	Markdown parser for lecture content conversion
Node.js	Latest	Runtime environment and package management

Build & Deployment:

- **Development Server:** Vite provides hot module replacement (HMR) for rapid development
- **Production Build:** Optimized bundle generation via Vite
- **Deployment:** GitHub Pages for static hosting and distribution
- **Package Manager:** npm for dependency management

3. Key Features & Modules

Photoelectric Effect Simulator

Interactive demonstration of the photoelectric effect allowing students to adjust light frequency and intensity, observe electron ejection, and explore the relationship between photon energy and work functions across different metals.

Black Body Radiation

Planck and Rayleigh-Jeans law comparison across wavelength and frequency modes. Illustrates the ultraviolet catastrophe with correct divergence rendering and a visible-spectrum gradient overlay. Part of the Failure of Classical Mechanics chapter.

Bohr Model Visualization

Visualizes electron orbits, energy level transitions, and spectral line emissions. Features an energy diagram with transitions, an inline Balmer series spectrum strip, and interactive controls for exploring hydrogen-like atoms.

Particle in a Box (1D & 2D)

Quantum wave function simulations showing energy quantization in infinite potential wells. 1D version shows wave patterns; 2D version includes contour plot visualization with superposition states and animation controls.

Quantum Tunneling & Barrier Scattering

Demonstrates particle scattering through finite potential barriers with visualization of reflection, transmission, and tunneling phenomena. Real-time animation of particle trajectories.

3D Atomic Orbitals

Interactive 3D probability density visualizations for hydrogen-like atoms using Three.js. Supports different quantum numbers (n , l , m_l) with isosurface rendering and prev/next orbital navigation.

Orbital Hybridization Visualizer

Interactive 3D visualization of sp , sp^2 , and sp^3 hybrid orbitals built with Three.js. Displays LCAO combination equations alongside the rendered orbitals and supports chip-based orbital selection. Files: `hybridization.ts`, `hybridization-math.ts`.

Diatomic Molecular Orbital (MO) Schemes

Tools for building and visualizing molecular orbital diagrams for diatomic molecules. Interactive diagram editor for determining bond properties and orbital interactions.

IR Vibrational Spectra

Simulates infrared absorption spectra for common molecules. Renders normal modes with wavenumber positions, relative intensities, and IR activity based on selection rules. Includes a characteristic band reference table (O-H, C-H, C=O, C-O). File: ir-spectra.ts.

Rotational Spectra

Microwave absorption simulation for diatomic molecules. Displays Boltzmann-weighted rotational populations, line positions from the rigid rotor model, and bond length determination from the rotational constant B . File: rot-spectra.ts.

Vibrational-Rotational Spectra

Realistic P, Q, and R branch structure for diatomic molecules with temperature and resolution controls. Uses molecule-specific B_0 and B_1 constants to model centrifugal distortion effects. File: vibrot-spectra.ts.

Integrated Lecture Materials

Repository of course lectures organized by week with PDF and Markdown formats. Supports mathematical notation via KaTeX for seamless integration with simulations. Extended through Week 12 including Lecture 34 secular equation supplement.

4. Project Architecture

Module Organization:

The project is organized into simulation modules, each with dedicated TypeScript files for logic and HTML entry points for UI. This modular design enables independent feature development and easy addition of new simulations.

Core Simulation Modules:

- **photoelectric.ts / photoelectric.html** - Photoelectric effect with graph.ts for visualization
- **blackbody.ts / blackbody.html** - Black body radiation with Planck and Rayleigh-Jeans comparison
- **bohr.ts / bohrSimulation.ts** - Bohr model with energyDiagram.ts for level diagrams
- **particleBox*.ts / particleBox*.html** - 1D and 2D quantum well simulations
- **barrier.ts / barrierSimulation.ts** - Quantum tunneling and scattering
- **atomic-orbitals.ts** - 3D orbital visualization with Three.js and orbitals-math.ts
- **hybridization.ts / hybridization-math.ts** - 3D hybrid orbital visualizer with LCAO equations
- **mo-scheme.ts / mo-diagram.ts** - Molecular orbital diagram creation and rendering
- **ir-spectra.ts / ir-spectra.html** - IR vibrational spectra with normal mode simulation
- **rot-spectra.ts / rot-spectra.html** - Rotational spectra and Boltzmann populations
- **vibrot-spectra.ts / vibrot-spectra.html** - Vibrational-rotational P/Q/R branch structure

Shared Infrastructure:

- **graph.ts**: 2D graphing utilities using Chart.js; colors wired to CSS tokens via getComputedStyle
- **simulation.ts**: Base simulation shell; canvas background reads --canvas-bg CSS token
- **transmissionGraph.ts**: Transmission coefficient rendering for barrier scattering
- **energyDiagram.ts**: Energy level diagram rendering
- **orbitals-math.ts**: Mathematical functions for hydrogen orbitals
- **hybridization-math.ts**: LCAO coefficient calculations for hybrid orbital construction
- **mo-diagram.ts**: Molecular orbital diagram logic
- **landing.ts**: Landing page with resume card, course progress strip, and TOC rendering
- **simulations.ts / simulations.html**: Simulations index page with chapter-organized TOC
- **lectures.ts / lecture-view.ts**: Lecture content management and display

Styling & UX:

The platform was redesigned in April 2026 from a glass-morphism aesthetic to an editorial-textbook visual language (Direction A v2). The new design system uses a paper/ink/accent CSS token palette, flat component geometry, and a consistent simulation shell applied across all pages. Full mobile responsiveness was added in the same cycle. See Section 5 for full design system details.

5. Visual Design System

In April 2026 the platform underwent a full visual redesign from a glass-morphism aesthetic to an editorial-textbook design language, designated Direction A v2. This work was co-developed with Claude Sonnet 4.6 across three coordinated commit phases.

Design Token Palette:

The redesign replaced ad-hoc hex values with a structured CSS custom property system:

- **--paper** (#f4f1ec): primary page background, replaces frosted-glass white
- **--ink**: primary text color for body content
- **--ink-3** (#7b7681): secondary/muted text; 3.92:1 contrast on --paper (AA large)
- **--accent**: interactive highlight color used on chips, borders, and active states
- **--canvas-bg**: dark simulation canvas background, replaces hardcoded #080c10
- **--plate-paper**: off-white used for the vintage spectral hero plate on the home page

New CSS Components:

Eight reusable component classes were introduced and applied uniformly across all simulation pages:

- **.breadcrumb**: hierarchical page location indicator below the nav bar
- **.page-header**: standardized simulation title block with description
- **.fig-caption / .fig-desc**: italic figure labels and descriptive captions below canvases
- **.readouts**: numeric output display panel for live simulation values
- **.qc-slider**: styled range inputs replacing browser defaults
- **.qc-chip**: pill-shaped selection buttons for molecule, metal, and axis options
- **.qc-prog**: course progress strip on the home page
- **.toc-row**: chapter-organized table-of-contents row used in simulations.html

Simulation Visual Upgrades:

Canvases across all simulations were updated to read colors from CSS tokens rather than hardcoded values, enabling consistent theming. Notable per-simulation changes include:

- **Bohr model**: dark grid canvas, amber nucleus (#ffb347) with glow ring, active orbit rendered amber/solid vs. white/dashed for inactive orbits, element label at bottom
- **IR spectra**: Chart.js dataset color updated to #ffd580 (golden); axis, grid, and label colors updated for dark canvas readability
- **MO diagram**: electron arrows switched to --ink token (previously invisible on light canvas)

Home Page Design Elements:

The index.html landing page received several editorial-style additions:

- **Vintage spectroscopic hero plate**: inline SVG showing H Balmer, Na D doublet, and Fe arc emission strips with wavelength annotations (H-beta highlighted at 486 nm)
- **Resume card**: white background panel with 3px accent left-border per Direction A spec
- **Course progress strip**: visual indicator of module completion, stored in localStorage
- **Einstein quote**: 1912 quotation placed at the page footer
- **Page tagline**: updated to 'from the first failures of classical mechanics to molecular orbitals'

Per-Page Figure Additions:

Scientific reference figures were added to individual simulation pages:

- **Bohr page:** Fig 4.2 Balmer series spectrum strip (inline SVG, four emission lines at 410, 434, 486, and 656 nm rendered with spectral colors)
- **Photoelectric page:** italic fig-desc below the canvas describing the experimental setup
- **IR spectra page:** Fig 3.2 characteristic band reference table covering O-H, C-H, C=O, and C-O regions in a flat white-panel grid layout

Mobile Responsiveness:

Full mobile responsiveness was added across all 15 pages. Layouts reflow for narrow viewports, navigation collapses to a hamburger menu, and canvas and chart sizes adapt to screen width.

6. Development History

The QuantumChem project has evolved significantly since its inception, growing from a basic photoelectric effect simulator into a comprehensive educational platform with multiple advanced simulations.

Key Development Milestones:

Phase 1: Foundation (Early Development)

- Initial implementation of photoelectric effect simulation
- Graph visualization system using Chart.js
- Basic UI framework and landing page
- Setup of TypeScript and Vite build system

Phase 2: Core Simulations (Bohr & Particle in a Box)

- Bohr model visualization with energy level diagrams
- 1D particle in a box quantum wave simulation
- Implementation of animation controls and parameter sliders
- Performance optimizations and code splitting

Phase 3: Advanced Features (Tunneling & Orbitals)

- Quantum tunneling through finite barriers
- 3D atomic orbital visualization with Three.js
- Integration of course lectures (Weeks 1-7+)
- Mathematical rendering with KaTeX

Phase 4: Extended Simulations (2D & Molecular Orbitals)

- 2D particle in a box with contour plot visualization
- Diatomic molecular orbital scheme builder
- Superposition state controls for particle in a box
- Enhanced diagram rendering and interaction

Phase 5: Hybridization & Molecular Spectroscopy

- Orbital hybridization visualizer with interactive 3D Three.js rendering
- IR vibrational spectra simulation with normal modes and selection rules
- Rotational spectra with Boltzmann populations and bond length determination
- Vibrational-rotational P/Q/R branch structure with temperature controls
- Black body radiation with Rayleigh-Jeans comparison and UV catastrophe rendering
- Chemistry axis convention alignment (Z vertical, Y horizontal, X out of plane)
- Lecture materials extended through Week 12 (Lecture 34 secular equation supplement)

Phase 6: Editorial-Textbook Redesign & Mobile (Current)

- Direction A v2 design system: paper/ink/accent CSS token palette
- Eight new CSS components applied across all 15 pages (breadcrumb, page-header, fig-caption, readouts, qc-slider, qc-chip, qc-prog, toc-row)
- Canvas colors wired to CSS tokens via getComputedStyle across all simulations
- Vintage spectroscopic SVG hero plate and Einstein quote on home page
- Bohr page: Balmer spectrum strip and amber nucleus with glow ring
- IR spectra page: characteristic band reference table
- Simulations hub reorganized to chapter-section TOC layout
- Full mobile responsiveness across all pages with hamburger navigation

Recent Commits (Latest 10):

- feat: add Lecture 34 supplement with complete secular equation derivation
- Add full mobile responsiveness to all pages
- Add Molecular Spectroscopy section with three new simulations
- Add spectral hero plate, fix blackbody graph rendering
- Add design_handoff_2 visual elements: spectrum strip, fig-descs, band table, Einstein quote
- Phase 2: simulation visual upgrades + sniff-check fixes
- Apply Direction A v2 editorial-textbook redesign (Phase 1)
- Upgrade hybridization visualizer to interactive 3D (Three.js)
- Add Orbital Hybridization Visualizer
- Add Black Body Radiation simulation to Failure of Classical Mechanics

This is an open-source educational project designed to democratize access to physics and chemistry simulations.

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